11.5 Let consider $\sum_{n=1}^{\infty}(-1)^n b_n$ is an alterbny series then:

If $\lim_{n\to\infty} b_n = 0$ and b_n is a decreasing sequence then

$$\sum_{n=1}^{\infty}(-1)^n b_n.$$

Example: Let consider $\sum_{n=1}^{\infty}\frac{(-1)^n}{n}$. this is a convergent series b/c.

$b_n = \frac{1}{n}$, $\lim_{n\to\infty} b_n = 0$ and $n+1 > n \Rightarrow \frac{1}{n} \geq \frac{1}{n+1}$

so b_n is a decreasing sequence then A-T

$$\sum_{n=1}^{\infty}\frac{(-1)^n}{n} < \infty.$$

Absolute Convergence and Conditional convergence.

Let consider series $\sum a_n$ and also $\sum |a_n|$

$$\sum_{n=1}^{\infty}|a_n| = |a_1| + |a_2| + |a_3| + \cdots$$

Def. A series $\sum a_n$ is absolutely convergent if $\sum |a_n|$ is convergent.

Example: Let $\sum_{n=1}^{\infty}\frac{(-1)^n}{n^3}$ and $\sum_{n=1}^{\infty}\left|\frac{(-1)^n}{n^3}\right| = \sum_{n=1}^{\infty}\frac{1}{n^3}$ is

convergent b/c it is a p-series $p = 3 > 1$. So, $\sum_{n=1}^{\infty}\frac{(-1)^n}{n^3}$ is absolutely convergent.

Def: A series $\sum a_n$ is conditionally convergent if it is convergent but $\sum |a_n|$ is divergent.

Example: We know that $\sum_{n=1}^{\infty}\frac{(-1)^n}{n}$ is convergent but

$$\sum_{n=1}^{\infty}\left|\frac{(-1)^n}{n}\right| = \sum_{n=1}^{\infty}\frac{1}{n} \text{ is divergent}$$

then, $\sum_{n=1}^{\infty}\frac{(-1)^n}{n}$ is conditionally convergent.

Thm. If a series $\sum a_n$ is absolutely convergent, then it is convergent.

Example: Determine whether the series

$$\sum_{n=1}^{\infty} \frac{\cos n}{n^2}$$

is convergent or divergent.

Sln. $$\sum_{n=1}^{\infty} \left| \frac{\cos n}{n^2} \right| = \sum_{n=1}^{\infty} \frac{|\cos n|}{n^2}.$$

Notice that

$$\frac{|\cos n|}{n^2} \le \frac{1}{n^2}$$

and $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent by p-series $p = 2 > 1$. and using Direct comparison thm $\sum_{n=1}^{\infty} \frac{|\cos n|}{n^2}$ is convergent.

So $\sum_{n=1}^{\infty} \frac{\cos n}{n^2}$ is absolutely convergent and by the thm it is convergent.

Problem 1. Determine whether the series is absolutely convergent, conditionally convergent, or divergent.

i). $\sum_{n=1}^{\infty} \frac{1 + 2\sin n}{n^3}$ ii). $\sum_{n=1}^{\infty} \frac{\cos n\pi}{3n+2}$

Sln: ii). $\sum_{n=1}^{\infty} \left| \frac{\cos n\pi}{3n+2} \right| = \sum_{n=1}^{\infty} \frac{|\cos n\pi|}{3n+2} = \sum_{n=1}^{\infty} \frac{1}{3n+2}$. this is divergent!

$$\sum_{n=1}^{\infty} \frac{\cos n\pi}{3n+2} = \sum_{n=1}^{\infty} \frac{(-1)^n}{3n+2}$$

By A-t $\sum_{n=1}^{\infty} \frac{\cos n\pi}{3n+2}$ is

$b_n = \frac{1}{3n+2}$: $\lim_{n\to\infty} b_n = 0$ ✓

$n+1 > n \Rightarrow 3(n+1) > 3n$

$\Rightarrow 3(n+1)+2 > 3n+2$

$\Rightarrow \frac{1}{3n+2} > \frac{1}{3(n+1)+2}$

convergent then $\sum_{n=1}^{\infty} \cos n\pi$ is conditionally convergent. $b_n > b_{n+1}$.

i) Case 1: $|1+2\sin n| \le 3$ then

$$\frac{|1+2\sin n|}{n^3} \le \frac{3}{n^3}$$

and $\sum_{n=1}^{\infty} \frac{3}{n^3} = 3\sum_{n=1}^{\infty} \frac{1}{n^3}$ is convergent b/c it is a p-series with $p=3>1$.

So, by Direct comparison thm $\sum_{n=1}^{\infty} \frac{|1+2\sin n|}{n^3}$ is convergent.

Hence, $\sum_{n=1}^{\infty} \frac{1+2\sin n}{n^3}$ is absolutely convergent and therefore

$$\sum_{n=1}^{\infty} \frac{1+2\sin n}{n^3} \text{ is convergent.}$$

Case 2. $\sum_{n=1}^{\infty} \frac{1}{n^3} + \sum_{n=1}^{\infty} \frac{2\sin n}{n^3} = \sum_{n=1}^{\infty} \frac{1+2\sin n}{n^3}$ if $\sum_{n=1}^{\infty} \frac{1}{n^3}$ and $\sum_{n=1}^{\infty} \frac{2\sin n}{n^3}$ is convergent. Indeed, $\sum_{n=1}^{\infty} \frac{1}{n^3}$ is convergent by p-series and $\sum_{n=1}^{\infty} \frac{2\sin n}{n^3} \le \sum_{n=1}^{\infty} \frac{2}{n^3}$ which is convergent as well

then $\sum_{n=1}^{\infty} \frac{1+2\sin n}{n^3}$ is convergent. is it absolutely convergent?

$$\frac{|1+2\sin n|}{n^3} \le \frac{|1| + 2|\sin n|}{n^3} = \frac{1}{n^3} + 2\frac{|\sin n|}{n^3}$$

and since $\sum_{n=1}^{\infty} \frac{1}{n^3} < \infty$ and $\sum_{n=1}^{\infty} 2\frac{|\sin n|}{n^3}$ is convergent

then $\sum_{n=1}^{\infty} \left(\frac{1}{n^3} + 2\frac{|\sin n|}{n^3}\right)$ is convergent by the comparison thm

$$\sum_{n=1}^{\infty} \frac{|1+2\sin n|}{n^3} \text{ is convergent.}$$

Hence $\sum_{n=1}^{\infty} \frac{1+2\sin n}{n^3}$ is absolutely convergent.